ACADEMIC & ENGINEERING SUBMISSION

GYM & FITNESS MANAGEMENT SYSTEM (GFMMS)

ANYTIME ATHLETIC (AF·24 PROTOCOL)

COMPREHENSIVE PROJECT REPORT & VERIFICATION DOSSIER

Project Domain:	**Enterprise Web Application & Biometric Access Control**
Core Modules:	**Module A (Registration & Membership) + Module B (Turnstile & Classes)**
Backend Architecture:	**TypeScript, Node.js, Express.js, PostgreSQL 17, Prisma ORM**
Security & Auth:	**1-Click Google OAuth, JWT, Bcrypt, Role-Based Access Control**
Hardware Simulator:	**html5-qrcode Webcam Optical Scanner + Web Audio Synth**
Notification Engine:	**Nodemailer Gmail SMTP (dynamitedhruvin@gmail.com)**
Automated Tests:	**13/13 Jest Tests Passing • 21/21 E2E REST Scenarios Passing**
Lead Engineer / Author:	**Dhruvin Patel**
Submission Date:	**October 24, 2026**
System Status:	**100% OPERATIONAL & VERIFIED (PASS)**

Anytime Athletic 24/7 Platform — Engineering Project Documentation

Verified against GFMMS Software Requirements Specification and UML Design Contracts

1. EXECUTIVE SUMMARY & ARCHITECTURE

The **Gym & Fitness Management System (GFMMS)** is an enterprise-grade full-stack architecture designed to power 24/7 athletic facilities. Operating under the high-performance brand **Anytime Athletic (AF·24 Protocol)**, the platform replaces fragmented gym spreadsheets with a synchronized, real-time reactive ecosystem.

> **Core Objectives Achieved:**
>
> - **Module A:** Comprehensive member registration, atomic payment verification, 4-tier plan lifecycle, and dynamic QR passcode generation.
> - **Module B:** Biometric turnstile simulation, 3-day emergency grace buffer, 60-minute anti-passback guard, FIFO class waitlist engine, and automatic 15-minute class attendance proximity matching.
> - **Advanced Integrations:** Real webcam QR scanner via `html5-qrcode`, procedural Web Audio laser sweep chirp, 1-Click Google OAuth, and 18% GST tax invoicing.

SYSTEM TECHNOLOGY STACK

Layer	Technology	Implementation Details
Runtime & Language	Node.js v20+ / TypeScript 5	Strict type safety, ES Modules, zero compilation errors (`tsc --noEmit`).
Application Server	Express.js 4	Modular routers, centralized error middleware, CORS, JSON payload handling.
Database & ORM	PostgreSQL 17 / Prisma ORM	Strict relational foreign keys, cascade constraints, native Enums.
Authentication	JWT + Bcrypt + Google OAuth	Dual-role RBAC (Owner vs Athlete) with instant 1-click account switching.
Optical Hardware Scan	html5-qrcode + Web Audio	Real-time webcam video stream QR decode with procedural audio feedback.
Transactional Email	Nodemailer (Gmail SMTP)	Live dispatch of welcome passes, 18% GST invoices, and owner entrance alerts.
Frontend Interface	Tailwind CSS 3 / Vanilla ES6	Responsive dark athletic aesthetic, multi-layer depth parallax, zero AI filler.
Automated Testing	Jest & Supertest	13 automated unit/integration tests + 21-scenario E2E system audit suite.

2. MODULE A: MEMBER REGISTRATION & MEMBERSHIP

Module A governs athlete enrollment, payment verification, and subscription state machines:

CORE BUSINESS RULES & OPERATIONS

- **Atomic Member Registration:** Enrolls athletes with full contact, date of birth, emergency phone, and selected plan.
- **Conditional State Machine:** Membership status remains `Pending` or `Cancelled` until an authorized payment is settled. Upon payment success, status immediately becomes `Active`.
- **QR Identifier Generation:** Every activated athlete receives a unique, human-readable turnstile pass token (e.g. `AF-QR-ALICE001`) and high-resolution optical QR code.
- **Payment Failure Integrity Guard:** If payment processing fails (simulated via `paymentMethod: 'FAIL'`), the membership state is set to `Cancelled` with HTTP 400 rejection and no active pass is provisioned.
- **4-Tier Membership Plans:** Configured with 1-Month Anytime Flex (₹2,499), 3-Month Transformation (₹6,499), 6-Month Elite (₹10,999), and 12-Month All-Access VIP (₹18,999).
- **30-Minute Expiration Policy:** Incomplete registrations expire automatically after 30 minutes, preventing orphaned allocations.

MODULE A ENDPOINTS REFERENCE

Method & Path	Description	Success Response
`POST /members/register`	Enrolls athlete and atomically charges subscription	`201 Created` with member, active membership & QR token
`GET /members`	Lists all registered members with active plans	`200 OK` with array of member objects
`GET /members/:id`	Retrieves complete athlete profile and history	`200 OK` with relational memberships, payments, attendances
`GET /members/:id/qrcode`	Generates high-res Base64 optical QR data URL	`200 OK` with Base64 PNG data URL
`GET /plans`	Fetches all active membership plans	`200 OK` with plan tiers, pricing, and durationDays

3. MODULE B: ATTENDANCE & CLASS BOOKING

Module B governs turnstile hardware control, access policies, and group studio reservations:

CORE BUSINESS RULES & OPERATIONS

- **Dual-Identifier Resolution:** The access controller resolves both database UUIDs and human-readable QR tokens (e.g. `AF-QR-ALICE001`).
- **Access Gatekeeper:**
  - **Active Membership:** Turnstile unlatches, attendance record created with timestamp and method (`QR` / `Manual`).
  - **3-Day Grace Buffer:** Subscriptions expired within 72 hours are permitted emergency access with an urgent renewal reminder.
  - **Expired Outside Grace:** Access rejected with HTTP 403 (`checkInDenied – please renew`); zero attendance rows created.
- **60-Minute Anti-Passback Guard:** Rejects duplicate check-ins within 60 minutes with HTTP 403 to prevent badge sharing.
- **15-Minute Class Proximity Auto-Attendance:** Checking in within 15 minutes of a scheduled fitness class auto-marks the attendee as `Attended` and links the attendance record.
- **Smart Capacity & FIFO Waitlist Engine:**
  - Open seat: Status `Booked`, capacity increments.
  - Full class: Status `Waitlisted` in strict chronological FIFO queue order.
  - Cancellation > 2 hours before class: Auto-promotes earliest waitlisted candidate to `Booked`.
  - Cancellation < 2 hours before class: Rejected by policy to protect studio schedule integrity.

MODULE B ENDPOINTS REFERENCE

Method & Path	Description	Success Response
`POST /attendance/checkin`	Turnstile scan verifying status, grace period & anti-passback	`201 Created` with attendance ID and door unlatch
`POST /attendance/checkout`	Turnstile exit stamping departure timestamp	`200 OK` with updated attendance record
`GET /classes`	Studio classes roster with live occupancy counts	`200 OK` with capacity, currentBookings, schedule
`POST /bookings`	Reserves studio class seat or places into FIFO waitlist	`201 Created` with Booked or Waitlisted status

Method & Path	Description	Success Response
`POST /bookings/:id/cancel`	Cancels booking & promotes next waitlisted candidate	`200 OK` with promotion trigger confirmation

4. ADVANCED EXTENSIONS & VERIFICATION EVIDENCE

1-CLICK GOOGLE OAUTH & RBAC

Provides passwordless authentication for both administrative owners and athletes. Integrated directly into the SPA header with quick demo profiles for Alice Johnson, Bob Smith, Charlie Davis, and Club Owner Dhruvin, plus support for any custom Google email address.

HARDWARE WEBCAM QR SCANNER & AUDIO SYNTHESIS

The entrance kiosk embeds a real-time webcam video feed powered by `html5-qrcode`. When a QR pass is detected, a procedural Web Audio synthesizer emits an authentic 988Hz-1976Hz laser chime, populating the input and triggering turnstile check-in automatically.

GST INVOICING & TRANSACTIONAL EMAILING

Every transaction generates a GST-compliant tax invoice calculating 18% tax breakdown, subtotal, total amount, and unique invoice code. Connected to Gmail SMTP (`dynamitedhruvin@gmail.com`), the backend delivers welcome packets with attached scannable QR passes and live owner entry alerts.

AUTOMATED VERIFICATION MATRIX

Suite	Commands Executed	Results	Status
Jest Unit & Integration	`npm test`	13 Tests Passed / 0 Failed	PASS
TypeScript Compiler	`npx tsc --noEmit`	0 Errors / Clean Build	PASS
E2E REST Audit Suite	`npx ts-node scripts/comprehensive-audit.ts`	21 Scenarios Passed / 0 Failed	PASS

5. OFFICIAL SCREENSHOT SUBMISSION GUIDE

The following guide details all 13 screenshots required for academic and project submission, including page location, capture frame, and the exact descriptive caption to print below each figure:

#	Figure Title	Page Location	Visual Elements to Capture	Exact Caption / Description
SS-01	**Hero Landing Page & Branding**	Dashboard (`/`)	Header, kinetic SVG crest, hero headline, trust badges.	*Figure 1: Anytime Athletic Landing Page featuring kinetic SVG branding and multi-layer parallax depth system.*
SS-02	**1-Click Google OAuth Modal**	Header → Sign In Modal	Quick profile cards for Alice, Bob, Charlie, and Owner.	*Figure 2: Unified 1-Click Google OAuth modal allowing passwordless authentication for both owners and athletes.*
SS-03	**24/7 VIP Digital QR Pass**	Header → QR Pass Modal	High-res scannable QR code, token `AF-QR-ALICE001`, valid Oct 2027.	*Figure 3: Scannable 24/7 VIP Digital Entrance Pass with high-resolution QR code, token passcode, and test kiosk action.*
SS-04	**Optical Camera Scanner**	Kiosk (`/kiosk`)	Live camera feed, sweeping red laser beam, sensor indicators.	*Figure 4: Hardware Turnstile Optical Camera Scanner simulating real-time video QR decoding with Web Audio chime synthesis.*
SS-05	**Turnstile Access Granted**	Kiosk after scan	Green neon banner: *"ACCESS GRANTED — WELCOME, ALICE JOHNSON!"*	*Figure 5: Successful Entrance Check-In confirmation validating Active membership and recording turnstile timestamp.*
SS-06	**60-Min Anti-Passback Guard**	Kiosk after re-scan	Red warning banner: *"ACCESS DENIED (Duplicate Check-In)"*	*Figure 6: Anti-Passback Security Enforcement rejecting a second check-in attempt within 60 minutes with HTTP 403 Forbidden.*
SS-07	**Enrollment & 4-Tier Plans**	Join Now (`/register`)	4 plan cards, enrollment form, payment method selector.	*Figure 7: Module A Membership Enrollment Protocol displaying 4 plan tiers, instant payment, and auto-renew toggle.*
SS-08	**Studio Classes & Capacity**	Classes (`/classes`)	Class cards with live capacity meters and booking buttons.	*Figure 8: Module B Studio Scheduling Engine showing real-time capacity meters, trainer assignments, and booking status.*

#	Figure Title	Page Location	Visual Elements to Capture	Exact Caption / Description
SS-09	**FIFO Waitlist Inspector**	Classes → Inspect Queue	Modal with ordered waitlist candidates for full class.	*Figure 9: FIFO Waitlist Management modal displaying queued athletes eligible for automatic promotion upon seat cancellation.*
SS-10	**Athlete Directory Cards Grid**	Members (`/members`)	3-column luxury athlete cards grid with avatars and badges.	*Figure 10: Active Athlete Directory in Card View mode showing member avatars, status rings, and quick action controls.*
SS-11	**Deep Athlete Dossier Modal**	Members → Click Dossier	Modal with tabs: Profile, Active Plan, Attendance, Invoices.	*Figure 11: Deep Athlete Dossier Modal displaying complete historical timeline including turnstile check-ins and invoice ledger.*
SS-12	**GST Tax Invoice Sheet**	Payments → Click Invoice	Official tax invoice sheet displaying 18% GST calculation.	*Figure 12: GST-Compliant Official Tax Invoice modal displaying 18% GST tax rate breakdown and verified transaction hash.*
SS-13	**Terminal Verification Output**	Terminal / PowerShell	Terminal displaying 21/21 PASS and 13/13 Jest tests.	*Figure 13: Execution evidence of the 21-scenario End-to-End System Audit Suite and 13-test Jest suite confirming zero errors.*